

CS/B.Tech/Even/IT/6th Sem/IT-603/2014

2014

Software Engineering

Time Alloted : 3 Hours

Full Marks : 70

*The figure in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable*

GROUP - A

(Multiple Choice Type Questions)

1. Choose the correct alternative for the following

10x1=10

- i) The type of failure that occurs for all input values while invoking a function of the system is
 - a) transient failure b) permanent failure
 - c) recoverable failure d) unrecoverable failure
- ii) When a system interfaces with other types of systems then that time the testing that will be required is:
 - a) volume testing b) configuration testing
 - c) compatibility testing d) maintenance testing
- iii) Beta Testing is done by
 - a) the developers
 - b) a select group of friendly customers

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- c) the test team
- d) the analyst
- iv) The chain of activities that determines the duration of the project is:
 - a) duration path
 - b) critical path
 - c) linearly independent path
 - d) none of these
- v) Which one of these is an example of algorithmic cost modelling?
 - a) COCOMO
 - b) McCabe cyclomatic complexity
 - c) MTTF
 - d) None of these.
- vi) Which is not a size measure for software?
 - a) LOC
 - b) Function Count
 - c) Cyclomatic complexity
 - d) Halsted's Program Length
- vii) Software failure rate is highest during
 - a) Design Phase
 - b) Testing Phase
 - c) Implementation Phase
 - d) Installation Phase
- viii) The basic premise of modern quality assurance is:
 - a) Continuous process improvement

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- b) thorough product testing
- c) if an organization's processes are good and are followed rigorously then the products are bound to be of good quality
- d) collection of process metrics
- ix) Project planning does not include
 - a) Risk identification
 - b) Design
 - c) Cost estimation
 - d) Configuration
- x) Big Bang Integration Testing is useful for projects with
 - a) Smaller no. of modules
 - b) Larger no. of modules
 - c) Average no. of modules
 - d) None of these

GROUP - B

(Short Answer Type Questions)

Answer any *three* of the following 3x5=15

- . What steps are required to perform SQA? What is the difference between Fault, Failure and Error?
- . What is Software Reliability? What are the different metrics involved in Software Reliability?
- . What are the different levels of Testing and their goals?
- . Define Jensen and Putnam's Model for project estimation.
- . Explain various types of Integration Testing.

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GROUP - C

(Long Answer Type Questions)

Answer any *three* of the following.

3x15=45

7. (a) Outline the structure of a Software Project Management Plan(SPMP) document.

(b) Consider a S/W development project planned for a small library. The activities, time requirements and precedence relations are given in the following table:

ACTIVITY	TIME(IN WEEKS)	PRECEDENCE
A	6	---
B	4	---
C	3	A
D	4	B
E	3	B
F	10	E
G	3	E,F
H	2	C,D

Draw the Gantt Chart and find out the critical path and duration of the entire Project.

[5+10]

8. (a) What are the differences between Black Box and White Box Testing?

(b) Distinguish between Software Verification and Software

Validation

(c) How will you represent CFG in the following cases:

- Sequence
- Selection
- Iteration

[5+5+5]

(a) How are Projects categorized in COCOMO? Why is COCOMO called Heuristic Estimation Technique?

(b) Explain the topic given below:

"When the schedule of a Project is compressed the required effort increases in proportion to the fourth power of the degree of the compression"

(c) Suppose a system for a office automation is to be designed. There are 4 major modules in the system: Data Entry, Data Update, Query and Report Generation. The Project falls in the organic category. The sizes of different modules are in the ratio 2:2:3:5. The Total size of the Project is 6KLOC in the 1st year and 12 KLOC in the 2nd year. The ratings of Cost Drivers are assumed to be as follows:

Complexity	High	1.5
Storage	High	1.06
Experience	High	1.13
Programmer Capability	High	1.17

Obtain Effort estimates for the entire Project for different Phases in Two Years.

[4+4+7]

10. (a) What is the difference between Code Walkthrough and Code Inspection.

(b) State McCabe's Cyclomatic Complexity.

(c) Find out McCabe's Cyclomatic complexity for the following program code by suitable methods:

```
int find_maximum(int i,int j,int k)
{
    int max=0;
    if (i > j) then
        if (i > k) then max=i;
        else max=k;
    else if (j > k) then max=j;
    else max=k;
    return(max);
}
```

[5+2+8]

11. Write short notes on any 3 of the following:

[3x5=15]

(a) McCall's Quality Triangle.

(b) Feasibility Study.

(c) RAD Model.

(d) Halstead's Metrics.

(e) Waterfall Model.